How new information affect Prob?

Prob is not fixed.

$\Rightarrow$ When new info (evidence) is available, our belief about an event changes

$\Rightarrow$ conditional probability

Ex Student $\Rightarrow$ passed

Attendance $\rightarrow$ 80%

$$P(A|B) = \frac{\text{Outcome where A and B happen}}{\text{Outcomes where B happens}}$$

100 students

60 $\rightarrow$ attended

Out of 60 $\rightarrow$ 45 passed

$P(\text{Pass} \mid \text{Attended})$

A is an event (student passed)

B is new evidence (student attended classes)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A)$ = Prior prob (before evidence)

$P(B)$ = Prob of the evidence

$P(A|B)$ = Posterior prob (after evidence)

belief updating

prior_pass = 0.5

likelihood = 0.8 # P(Attendance | Pass)

evidence = 0.6 # P (attendance)

posterior = (likelihood x prior_pass) / evidence

Bayes' Theorem

Reverse cond^n prob

$P(A|B) \times$

$$P(B|A) = \frac{P(A|B) * P(B)}{P(A)}$$

or

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$P(A)$ = Prior prob

$P(B|A)$ = likelihood

$P(B)$ = Evidence

$P(A|B)$ = Posterior prob

$\Rightarrow$ A student attended classes
What is the prob they passed? } Cond^n

$\Rightarrow$ A student passed the exam
What is the prob they attended the classes? } Bayes' Theorem

https://colab.research.google.com/drive/1JpGuGQQrUkpgx5UiGvrhknx7Y5PND6WR?usp=sharing

$P(D) = 0.02$

$P(+ \mid D) = 0.95$

$$P(+) = (0.95 \times 0.02) + (0.05 \times 0.98)$$

$$= 0.068$$

$$P(D \mid +) = \frac{0.95 \times 0.02}{0.068}$$

$$= 0.27$$

```
import pandas as pd

df=pd.DataFrame({
    "attendance": ["yes","no","yes","no","yes","no"]
    , "result": ["pass","fail","pass","fail","pass","fail"]
})

prob_pass=(df["result"]=="pass").mean()

print(prob_pass)

# filter the students wjo attended the classes

attended=df[df["attendance"]=="yes"]

print(attended)

prob_pass_given_attendance=(attended["result"]=="pass").mean()

print(prob_pass_given_attendance)
```

```
0.5
  attendance result
0        yes   pass
2        yes   pass
4        yes   pass
1.0
```

```
A disease affects 1% of people.
● Test is 90% accurate
● A person tests positive
What is the probability the person actually has the disease?
```

```
P_disease=0.01
P_positive_given_disease=0.9
P_positive=0.1
P_disease_given_positive = (P_positive_given_disease * P_disease) / P_positive

print(P_disease_given_positive)
```

```
0.09000000000000001
```

```
Based on past data:
● 70% of students who revise daily pass
● 40% of all students revise daily
● 60% overall pass rate

Find: P(Revise Daily | Pass)
```

```
P_pass=0.6
P_pass_given_revise=0.7
P_revise=0.4

# Bayes formula
#P(revise_daily | pass)
P_revise_given_pass=(P_pass_given_revise * P_revise) / P_pass

print(P_revise_given_pass)
```

```
0.4666666666666666
```

```
'''
A teacher wants to understand how attendance affects
exam results.
From past experience, the teacher believes that:
● A student passes the exam 50% of the time.
This is the prior probability.

The teacher also knows the following behavior:
● If a student passes, they attend classes 80% of the
time.
●If a student fails, they attend classes 40% of the
time.
'''
```

```
import random
import pandas as pd

# step 1: simmulate students
students=[]

for i in range(1000):
  # prior belief: P(pass)
  passed=random.random() < 0.5

  # likelihood
  # if student passes -> attends with 80% prob
  # if student fails -> attens with 40% prob

  if passed:
    attended= random.random() < 0.8
  else:
    attended=random.random() <0.4

  students.append({"pass":passed,"attend": attended })


# create dataframe
df=pd.DataFrame(students)
print(df)


# step 2: prior prob
prior = df["pass"].mean()
print(prior)

# step 3: observe the evidence (attendance)
attended_students=df[df["attend"]==True]


# step 4: posterior prob
post=attended_students["pass"].mean()

print(post)
```

```
      pass  attend
0    False    True
1    False   False
2     True   False
3    False    True
4    False   False
..     ...     ...
995  False    True
996  False   False
997   True    True
998  False   False
999   True    True

[1000 rows x 2 columns]
0.495
0.6699507389162561
```

```
'''
A medical test is used to detect a rare disease.
● 2% of people have the disease.
● If a person has the disease, the test is positive 95% of the time.
● If a person does not have the disease, the test isnpositive 5% of the time.
A patient tests positive.
```

```
Find:
1. Prior probability of disease
2. Probability that the patient actually has the
disease
3. Verify using a Python experiment
```

```
import pandas as pd
import random

data=[]

for _ in range(5000):
  disease=random.random() <0.02

  if disease:
    test = random.random() <0.95
  else:
    test= random.random() <0.05

  data.append({
      "disease": disease, "test_positive":test
  })

df=pd.DataFrame(data)

print(df[df["test_positive"]== True]["disease"].mean())
```

```
0.2672413793103448
```

```
QUESTION 3: Spam Email Detection
An email system tracks spam behavior:
● 30% of emails are spam.
● 90% of spam emails contain suspicious keywords.
●20% of non-spam emails contain suspicious keywords.
An email contains suspicious keywords.
Find the probability that it is spam.
```

```
QUESTION 4: Interview Selection & Skills
In a company hiring process:
● 40% of candidates are skilled
● Skilled candidates clear interviews 85% of the time
● Unskilled candidates clear interviews 30% of the time
A candidate has cleared the interview.
Find the probability that the candidate is skilled.
```

```
QUESTION 5: Product Defect Detection (Manufacturing)
In a factory:
● 5% of products are defective
● A machine flags:\
92% of defective products
8% of non-defective products


A product is flagged.
Find the probability it is defective.
```

Double-click (or enter) to edit